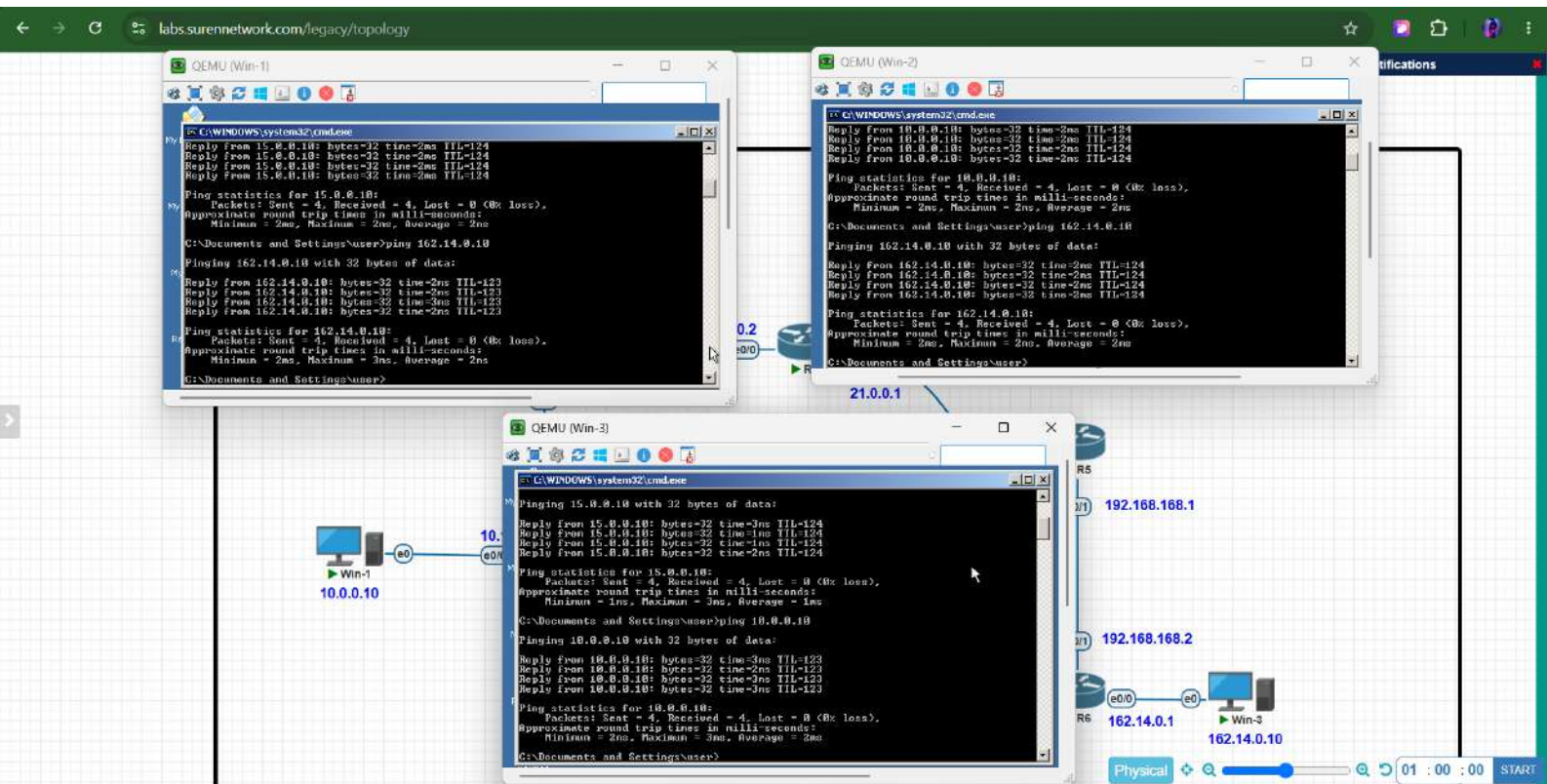


PROJECT LAB-1

Ping from PC's



Route config for R1, R2, R3

R1 Configuration:

```
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   10.0.0.0/8 is directly connected, Ethernet0/0
L   10.0.0.1/32 is directly connected, Ethernet0/0
S   15.0.0.0/8 [1/0] via 20.0.0.2
C   20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   20.0.0.0/8 is directly connected, Ethernet0/1
L   20.0.0.1/32 is directly connected, Ethernet0/1
S   21.0.0.0/8 [1/0] via 20.0.0.2
S   30.0.0.0/8 [1/0] via 20.0.0.2
S   162.14.0.0/16 [1/0] via 20.0.0.2
S   172.19.0.0/16 [1/0] via 20.0.0.2
S   192.168.168.0/24 [1/0] via 20.0.0.2

R1#
*Jul 21 02:01:02.943: %PNP-G-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1#
```

R2 Configuration:

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 [1/0] via 20.0.0.1
S   15.0.0.0/8 [1/0] via 30.0.0.2
C   20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   20.0.0.0/8 is directly connected, Ethernet0/0
L   20.0.0.2/32 is directly connected, Ethernet0/0
S   21.0.0.0/8 [1/0] via 30.0.0.2
C   30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   30.0.0.0/8 is directly connected, Ethernet0/1
L   30.0.0.1/32 is directly connected, Ethernet0/1
S   162.14.0.0/16 [1/0] via 30.0.0.2
S   172.19.0.0/16 [1/0] via 30.0.0.2
S   192.168.168.0/24 [1/0] via 30.0.0.2

R2#
```

R3 Configuration:

```
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 [1/0] via 30.0.0.1
S   15.0.0.0/8 [1/0] via 172.19.0.2
S   20.0.0.0/8 [1/0] via 30.0.0.1
C   21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   21.0.0.0/8 is directly connected, Ethernet0/2
L   21.0.0.1/32 is directly connected, Ethernet0/2
C   30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   30.0.0.0/8 is directly connected, Ethernet0/0
L   30.0.0.2/32 is directly connected, Ethernet0/0
S   162.14.0.0/16 [1/0] via 21.0.0.2
C   172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.19.0.0/16 is directly connected, Ethernet0/1
L   172.19.0.1/32 is directly connected, Ethernet0/1
S   192.168.168.0/24 [1/0] via 21.0.0.2

R3#
```

Route config for R4, R5, R6

The image displays three terminal windows showing the configuration for routers R4, R5, and R6 in a Packet Tracer environment. The configuration includes OSPF, EIGRP, and static routes.

R4 Configuration:

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 172.19.0.1
S 15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 15.0.0.0/8 is directly connected, Ethernet0/0
L 15.0.0.1/32 is directly connected, Ethernet0/0
S 20.0.0.0/8 [1/0] via 172.19.0.1
S 21.0.0.0/8 [1/0] via 172.19.0.1
S 162.14.0.0/16 [1/0] via 172.19.0.1
S 172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
L 172.19.0.2/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 172.19.0.1
R4#
```

R5 Configuration:

```
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 21.0.0.1
S 15.0.0.0/8 [1/0] via 21.0.0.1
S 20.0.0.0/8 [1/0] via 21.0.0.1
S 21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 21.0.0.0/8 is directly connected, Ethernet0/0
L 21.0.0.2/32 is directly connected, Ethernet0/0
S 30.0.0.0/8 [1/0] via 21.0.0.1
S 162.14.0.0/16 [1/0] via 192.168.168.2
S 172.19.0.0/16 [1/0] via 21.0.0.1
S 192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.1/32 is directly connected, Ethernet0/1
R5#
```

R6 Configuration:

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 192.168.168.1
S 15.0.0.0/8 [1/0] via 192.168.168.1
S 20.0.0.0/8 [1/0] via 192.168.168.1
S 21.0.0.0/8 [1/0] via 192.168.168.1
S 30.0.0.0/8 [1/0] via 192.168.168.1
S 162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 162.14.0.0/16 is directly connected, Ethernet0/0
L 162.14.0.1/32 is directly connected, Ethernet0/0
S 172.14.0.0/16 [1/0] via 192.168.168.1
S 192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.2/32 is directly connected, Ethernet0/1
R6#
```

The network diagram shows three routers (R4, R5, R6) connected to a central hub (R5). R4 is connected to R5 via Ethernet0/0 (10.0.10). R5 is connected to R6 via Ethernet0/0 (162.14.0.10). R5 is also connected to a central hub (R5) via Ethernet0/0 (162.14.0.10). The diagram includes a legend for Win-1 (10.0.10) and Win-3 (162.14.0.10).